

What is Minikube?

Minikube is a lightweight **Kubernetes** implementation that runs a single-node cluster on your local machine. It is mainly used for **development, testing, and learning** Kubernetes without needing a full multi-node cluster.

Key Features of Minikube

- ✓ **Runs Locally** – Ideal for development and testing.
- ✓ **Supports Multiple Hypervisors** – Works with VirtualBox, KVM, Hyper-V, Docker, etc.
- ✓ **Built-in Add-ons** – Includes useful services like dashboard, ingress, and metrics-server.
- ✓ **Fast Setup** – Quickly spins up a Kubernetes cluster.
- ✓ **Supports Load Balancers & Ingress** – Simulates real-world scenarios.

Start Minikube

```
minikube start
```

Check the status

```
minikube status
```

Deploy a sample app

```
kubectl create deployment hello-world --image=k8s.gcr.io/echoserver:1.10
```

Expose the app

```
kubectl expose deployment hello-world --type=NodePort --port=8080
```

Get the app's URL

```
minikube service hello-world --url
```

Stop Minikube

```
minikube stop
```

```
# Delete the cluster
```

```
minikube delete
```

```
minikube start
```

```
minikube status
```

```
kubectl get pod
```

```
kubectl get deploy
```

```
kubectl get replica
```

```
kubectl get pod -o wide
```

```
sudo apt install fontconfig openjdk-17-jre
```

```
java -version
```

```
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
```

```
https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
```

```
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]" \
```

```
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
```

```
/etc/apt/sources.list.d/jenkins.list > /dev/null
```

```
sudo apt-get update
```

```
sudo apt-get install Jenkins
```

```
sudo service Jenkins restart
```

```
sudo service Jenkins status
```

```
# sudo apt install docker.io -y
```

```
# sudo service docker restart
```

```
# sudo service docker status
```

```
# sudo usermod -aG docker $USER
```

```
docker images
```

```
docker ps
```

```
# sudo chmod 666 /var/run/docker.sock
```

```
sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
```

```
chmod +x kubectl
```

```
mkdir -p ~/.local/bin
```

```
mv ./kubectl ~/.local/bin/kubectl
```

```
kubectl version --client
```

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```
curl -LO https://github.com/kubernetes/minikube/releases/latest/download/minikube-linux-amd64
```

```
sudo install minikube-linux-amd64 /usr/local/bin/minikube && rm minikube-linux-amd64
```